

get around this problem, but they end up introducing the distracting halo effect in the darker scenes.

No Backlight Bleed

Backlight bleed, as the name suggests, is a typical problem faced by transmissive displays such as LCDs and LED-backlit LCDs. Conspicuous bright and dark spots spread randomly across the panel are tell-tale signs of this phenomenon affecting virtually all LED televisions. Such bleeding affects colour and picture uniformity, and is caused by range of factors ranging from imperfect backlight diffusion panels to chassis deformation. Since backlight bleed is linked to material and build quality, it is less prevalent in high-end LED panels. But because it affects cheaper displays the worst, it makes their already poor grey-scale performance end up looking even worse off. OLED TVs, on the other hand, don't suffer from these gremlins. These emissive panels don't have to bother with backlights and complicated diffusion panels like LCD panels, thereby making them completely immune to these issues.

Faster Response Time

LCD displays have a much slower pixel response time as compared to OLEDs, which are ten times as fast as conventional LED displays. Pixel response time represents the delay in transitioning from grey to black and white to black states by individual pixels. LEDs are marked by slow response times because the liquid crystals are reconfigured by electrical stimulation to modulate the backlight. The pixels are slow to refresh because the liquid crystals can't change states fast enough, thereby causing fast moving images to appear blurred to the human eye. Slow response time can be an issue if you watch a lot of sports, action movies, or play video games. Spend more on LED televisions and you'll find fancier models that employ electronic wizardry such as backlight scanning, higher refresh rates, frame interpolation, and pixel overdrive to mitigate the issue. Interestingly, these can also be applied to OLEDs for even better blur reduction.

Better Viewing Angles

Compared to OLED panels, LEDs are significantly more complicated. This complex design also results in the optimal viewing angle being restricted to a narrow sweet spot directly in front of the display. That's because transmissive LCD panels pass light through various layers of